Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/31/2026
PI Name:	Kim, Han Ph.D.		
Project Title:	Crew Population Characteristics: Perspectives for Spacesuit Fit		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Dynamic Loads**:Risk of In-Mission Injury and Performance Decrements and Long-term Health Effects due to Dynamic Loads		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	03/01/2025	**End Date:**	05/31/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/31/2026 per C. LaBelle/JSC (Ed., 11/19/25).		
Key Personnel Changes/Previous PI:	June 2026 update: Raj Prabhu, Ph.D. left the project. Chris Miller joined the project as Engineer (Ed., 6/21/26).		
COI Name (Institution):	Vining, Cassie (Aegis Aerospace/NASA Johnson Space Center) Newby, Nathaniel M.Sc. (KBR/NASA Johnson Space Center) Miller, Chris (KBR, Inc.)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Spacesuit-to-body interactions play a critical role in crew injury, mobility, and performance, especially during dynamic impact events or tasks requiring strenuous effort. Building spacesuits of which the shape and size match with the crewmember population is one of the primary goals in spacesuit design. However, some population subgroups, for example, small females or persons with uncommon body ratios, are still known to be less optimally accommodated. Thus, a structural assessment of crew population characteristics is required for suit fit, which can also help to develop guidelines for new suit designs by identifying the geometric features that can potentially mitigate fit issues.		

Rationale for HRP Directed Research:	Crew population characteristics information is urgently needed given the timeline of the System Capability Leadership Team (SCLT) Roadmap, thus insufficient time for full solicitation. Research involves spacesuit designs and CAD drawings protected under export licensing (ITAR/EAR). The use of suit fit and body scan data of NASA test subjects and crewmembers may require extensive review and approval processes for sharing with institutes outside of NASA. Research also requires guided interpretations of the data by extravehicular activity (EVA) subject matter experts (SMEs) and the use of documents under Controlled Unclassified Information (CUI) or export control, which may not be easily accessible from external institutes.
Research Impact/Earth Benefits:	This research replaces outdated, linear sizing methods with a data-driven 3D fitting framework to advance human-system integration on Earth across three major areas: 1. Occupational Safety and Protective Equipment: This technology directly improves the design of military body armor and first-responder gear, which historically fails to optimally fit female personnel and non-standard body shapes. By integrating actual 3D body-shape data, manufacturers can design custom-contoured protective systems that significantly enhance coverage, comfort, and field safety for a diverse workforce. 2. Automotive and Vehicle Ergonomics: The CAD-based human-modeling techniques developed in this study apply directly to vehicle cabin and seat design. This allows automotive and aviation engineers to optimize seating interfaces and safety restraints to safely accommodate a wider, more realistic range of the human population, reducing fatigue and collision injuries. 3. Sustainable Consumer Apparel: By replacing rigid, standard sizing tables with a 3D-driven custom-fitting workflow, the consumer apparel industry can transition to highly accurate, personalized clothing. This innovation will drastically reduce retail return rates, minimize material manufacturing waste, and support eco-friendly, on-demand production models.
Task Progress:	During the study period, this project successfully completed three primary aims: developing a predictive spacesuit fit model, designing a generative custom suit shape tool, and validating these models through human-in-the-loop evaluations and population-level projections. The work successfully defined spacesuit sizing from traditional linear measurements to quantitative evidence-based 3D geometry optimization. Phase 1: 3D Scanning and Virtual Fit-Modeling In this phase, the geometric relationship between the human body and the spacesuit hard upper torso (HUT) assembly was investigated. • Digitizing suit hardware: The team performed high-resolution 3D laser scans of the interior surfaces of the three standard Shuttle Extravehicular Mobility Unit (EMU) HUT sizes (Medium, Large, and Extra-Large). These scans were processed and converted into smooth, homologically-represented meshes. • Participant database integration: The team analyzed anonymized 3D body scans of suit fit-check participants to build the baseline training dataset. • Virtual suit alignment: An optimization algorithm was developed to virtually align the 3D body scans inside the 3D suit models. The algorithm positioned the body within the suit to minimize contact areas, and estimated spatial clearances. • Predictive sizing model: By linking these virtual 3D models with historical suit fit records, the team built a predictive fit-decision model. The complex 3D data was dimensionally reduced using principal component analysis (PCA). A machine learning model was then trained to predict exactly how a spacesuit will touch and fit any unique 3D body shape. Phase 2: Generative Design and Virtual Custom Suit Sizing Building upon the predictive model, the team developed an inverse design system capable of automatically morphing suit geometry to match an individual's body. • Generative sizing algorithm: The team developed an iterative design algorithm. When given a person’s 3D body scan, the algorithm starts with a standard suit size and automatically deforms its shape until it achieves the predicted optimal balance of contact support and clearance. Phase 3: Hardware Fabrication and Human-in-the-Loop Validation To validate the accuracy of the computer model's fit predictions, human-in-the-loop (HITL) testing was executed using 3D-printed mockup HUT. • Human subject tests with mockup HUT: Six human subjects were recruited to test the configured suits (with five completing the full testing suite). Subjects performed a standardized series of reaching, bending, and shoulder-mobility tasks. • Data collection: The team collected comprehensive physical data, including motion-capture tracking of joint movements and laser scans of the subjects wearing the suits to measure the exact alignment (indexing) of the hardware on the body. • Subjective fit mapping: To capture user comfort, subjects used interactive 3D authoring tools to color-code digital body models, identifying exact regions of comfortable contact, painful pressure, or excessive clearance. Phase 4: Key Biomechanical and Sizing Discoveries The physical testing and subjective feedback revealed critical details about spacesuit biomechanics. • Impact of HUT width (Q-distance): Wider spacing between the suit's shoulder joints (typical of oversized configurations) placed the subject's physical shoulder joints inside the suit's boundaries. This restricted their arm reach and caused severe underarm discomfort. Custom, narrower designs allowed for wider, smoother reach envelopes and eliminated joint pain. • Preference for tighter fit HUT: Tighter-fitting custom shapes distributed forces more evenly, avoiding painful localized pressure points and giving subjects a greater sense of physical control and boundaries. However, configurations that were too tight restricted full inhalation breathing. • Subject sizing preferences: When choosing their preferred fit, three subjects chose the tighter configuration (-1 size), one preferred the nominal custom fit, and one's preference depended on the task environment. None of the subjects preferred the oversized (+1) configurations, rating them as uncomfortable and highly restrictive. Phase 5: Population-Level Accommodation Projections

	The final phase of the project projected the validated model onto a larger scale to evaluate fleet-level sizing requirements. • Database projections: The model assessed a database of over 1,000 virtual human profiles representing the target crew population defined by NASA Human System Requirements. • Quantifying accommodation gaps: The analysis showed that the current three-size spacesuit inventory accommodates only a limited proportion of the crew population under optimal fit conditions, with a significant accommodation gap for women and smaller-statured individuals. • Sizing optimization: Sizing simulations demonstrated that expanding the standard sizing inventory would substantially increase the overall population accommodation rate. Phase 6: Software Tool Development and Handover The final technical output of the project was the integration of the fit models into a Graphical User Interface (GUI) design assistant tool built on the open-source program. This tool allows engineers to import a 3D body scan or enter manual dimensions, visualize the resulting touch and clearance gaps, and generate custom spacesuit geometries. These 3D custom designs can be exported in standard formats for direct use in engineering CAD software, ensuring the tools and methods are ready for integration into future NASA and commercial spaceflight programs.
Bibliography Type:	Description: (Last Updated: 06/22/2026)
Abstracts for Journals and Proceedings	Vining C, Kim H, Miller C, Newby N. "Crew population characteristics: Perspectives for spacesuit fit." 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. Abstracts. 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. , Apr-2026